

Miles Apart

Transportation Access and Economic Opportunity
Across Tennessee

Michele Howse | Nashville Software School | DA-17 | 2026

Demo Day — August 27, 2026

My Unique Perspective



Lyft Driver

I drive for Lyft in Nashville. I see firsthand which neighborhoods generate ride requests and which do not. For many riders, a rideshare trip is not a convenience; it is a necessity.



Insurance Analytics

My background at Professional Account Services, Inc., taught me how geography and access intersect with outcomes across 106 hospitals in 18 states.



Data Analyst

NSS DA-17 graduate with skills in Python, SQL, Tableau, Power BI, and Excel, so I am applying analytics to answer real questions about real communities.

The Problem

Tennessee is a state of striking contrasts. Nashville and Memphis are growing metro areas with expanding economies, while many rural counties face persistent poverty, limited public transit, and high rates of households without a vehicle.

Transportation gaps are not just inconveniences. They are barriers to employment, healthcare, and economic mobility.

95

Tennessee
Counties Analyzed

2

Data Sources
ACS + NTD

4

Data Questions
Answered

27

Transit Agencies
in Tennessee

Four Data Questions

1

How does car ownership vary across Tennessee counties, and how does it correlate with household income?

2

Which Tennessee counties have the least access to public transportation, and what are the characteristics of those communities?

3

How do commute methods differ between urban and rural Tennessee counties?

4

What is the relationship between transportation access and poverty or employment rates across Tennessee counties?

Data & Methodology

American Community Survey (ACS)

U.S. Census Bureau — 5-Year Estimates 2020–2024

- B08201 — Household Vehicle Availability
- B08301 — Means of Transportation to Work
- B19013 — Median Household Income
- S1701 — Poverty Status by County

National Transit Database (NTD)

Federal Transit Administration — 2024 Annual Data

- 27 Tennessee transit agencies identified
- Annual vehicle revenue hours by agency
- Unlinked passenger trips (ridership)
- Aggregated across all transit modes

Technologies Used:

Python / Pandas

SQL / SQLAlchemy

Tableau

PowerPoint

Conclusions & Takeaways



Car ownership and income are strongly linked. Counties with lower incomes consistently show higher rates of households without a vehicle.



Public transit is concentrated in a handful of urban counties. Most rural Tennessee counties have little to no transit coverage.



Urban and rural counties show very different commute patterns, but even urban counties rely heavily on solo driving.



Transportation access and poverty are deeply intertwined. The counties that need transportation options most are the ones with the fewest.

What Does This Mean?

Transportation is not just a transportation issue.

For Policymakers

Counties with high poverty and low vehicle ownership are the most underserved by public transit, and they need it most. Investment should be directed where access gaps are greatest.

For Employers

Transportation barriers affect workforce availability. Understanding which counties have limited access can inform hiring, remote work, and shuttle program decisions.

For Communities

The data shows that transportation access is a key factor in economic mobility. Addressing these gaps is essential for equitable economic development across Tennessee.

Thank You

Miles Apart: Transportation Access and Economic Opportunity Across Tennessee

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Questions?